Final Project Waveforms

mcu_tb.v

t=20 ns, play still low, so everything in reset condition; song=0, mode=0 (normal)

next_button high, song proceeds to #1

next_button goes high, song proceeds to #3

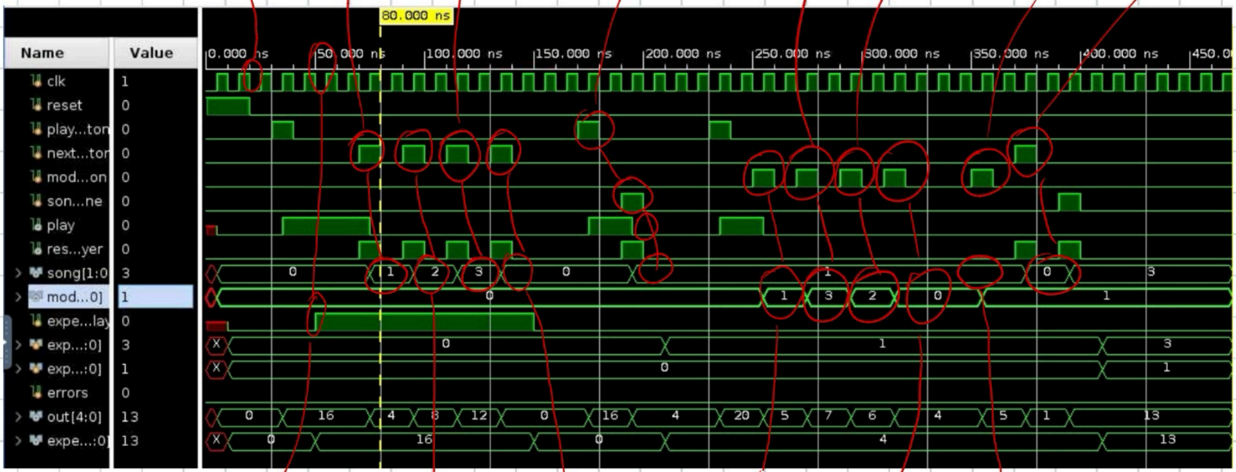
here play is song_done is pulsed, so song proceeds to #1 with paused condition

mode_button pressed, mode changes to rewind

mode changes to generate (for NN)

mode button changes mode to rewind

next button clicked, since we are on rewind mode, song changes from #1 → #0



play goes high, as expected, song is played with normal mode (0) and song 0

next_button high, song proceeds to #2

next_button high, song wraps around to #0

mode_button pressed, mode changes to #1 (fast forward)

after mode button clicked 4 times, mode wraps around to normal

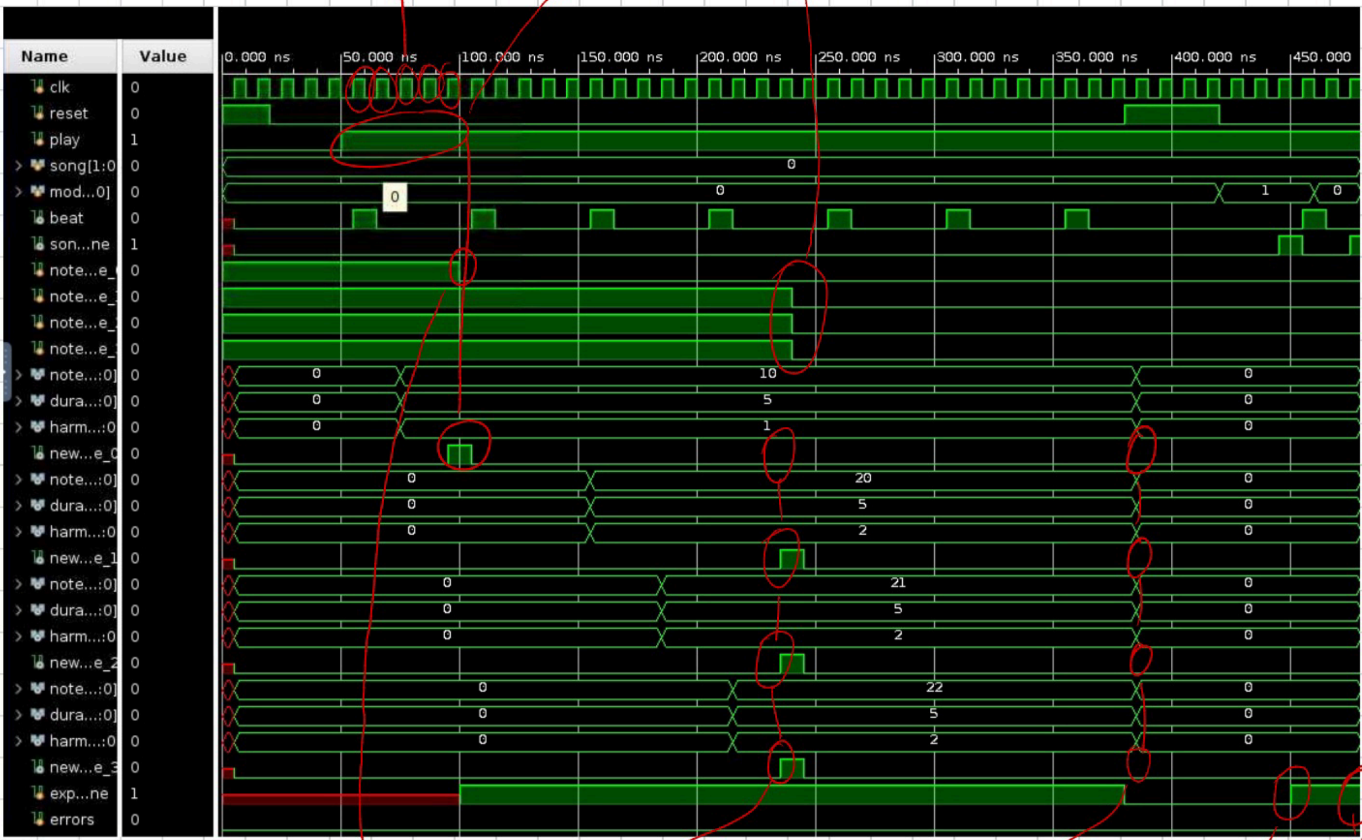
pressing mode button does not change song

song_reader_tb.v

play=1, so the FSM
transitions from
FETCH → DECODE → INC → FETCH ⇒ DECODE

after the 5-state trans,
new_note_0 is pulsed

now we make all
players busy,
notes should
be dropped



note_done_0 = 0, so
that player should
be busy

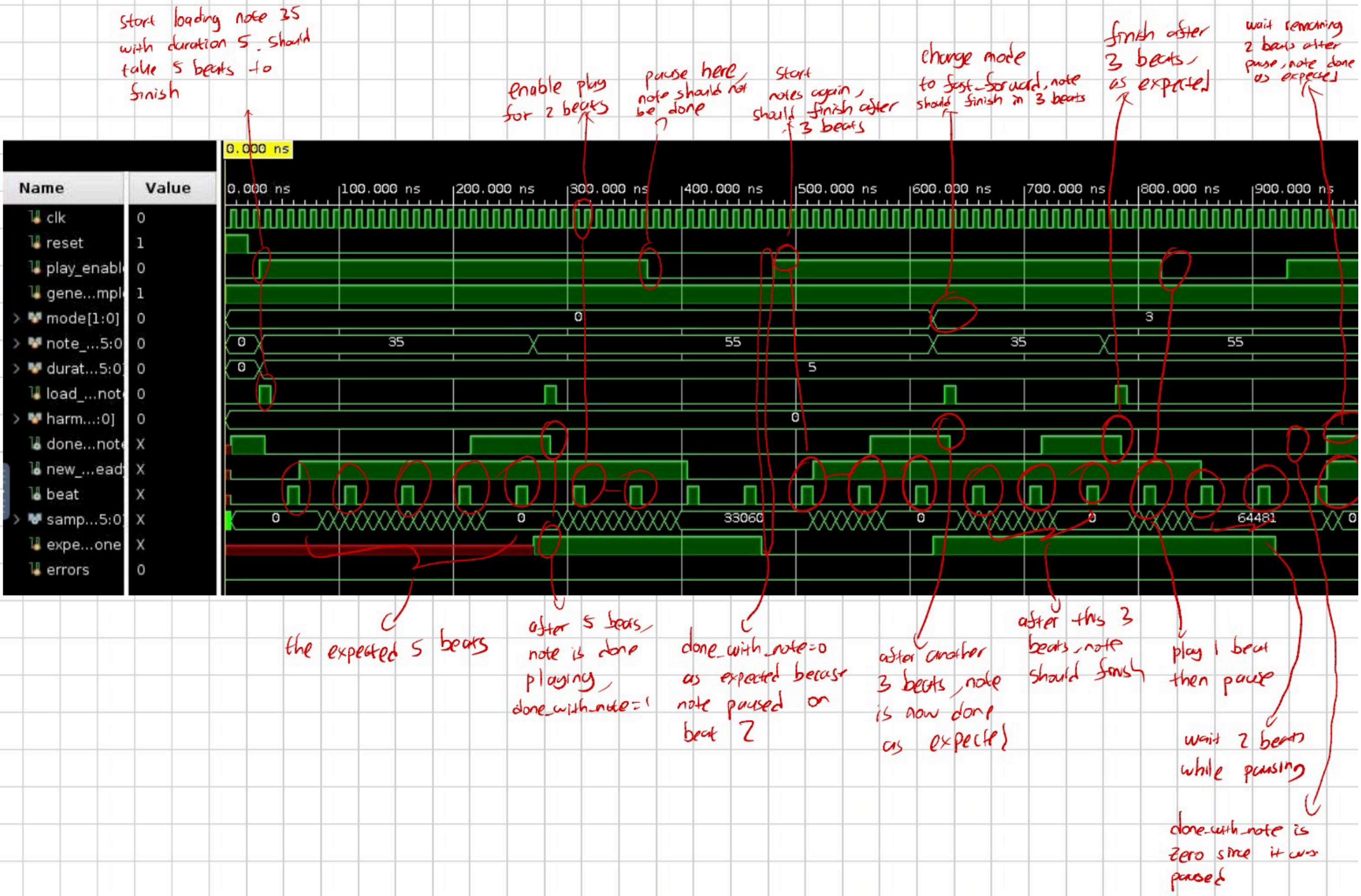
new_note 1,2,3 go high
as expected, since 0
is currently busy

here all new_notes
are zero, so all
notes are dropped

wait 3 cycles
after rewind,
hit INC state
so expected_done = 1

hit INC state
on NORMAL
mode, so
expected_done = 1

note_player_tb.v



sine_reader_tb.v

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1. quadrant 1, floating point addition rounded down
take two steps, expected_addr = 256(.50)
sample = 12539, expected = 12539

2. quadrant 1, floating point addition results in an extra address shift
take two steps, expected_addr = 256.50 + 256.50 = 513(.00)
sample = 23205, expected = 23205

3. quadrant 2, sample is flipped horizontally
take four steps, expected_addr = 513.00 + 513.00 = 1026(.00) --> 2(.00)
since this is quadrant 2, flip horizontally
expected_addr = 1023 - 2.00 = 1021(.00)
sample = 32767, expected = 32767

4. quadrant 3, sample is flipped vertically
take eight steps, expected_addr = 1026.00 - 1021.00 - 1 = 4(.00)
since this is quadrant 3, flip vertically
sample =   -201, expected =        -201

5. quadrant 4, sample is flipped horizontally and vertically
take eight steps, expected_addr = 4.00 + 1026.00 = 1030(.00) --> 6(.00)
since this is quadrant 4, flip horizontally and vertically
expected_addr = 1023 - 6.00 = 1017(.00)
sample = -32765, expected =      -32765

6. back to quadrant 1
take eight steps, expected_addr = 1026.00 - 1017.00 - 1 = 8(.00)
sample =    402, expected =    402

7. off, shift 1 step_size but not played
take eight steps, but it's paused
expected_addr = 8.00 + 128.25 = 136(.25)
sample =  6786, expected =  6786

8. test REWIND functionality: resumes from same point at sample
take one step in reverse
expected_addr = 136.25 - 128.25 = 8(.00)
sample =    402, expected =    402

9. test REWIND functionality: continuous sine wave
take eight steps in reverse
expected_addr = 8.00 - 1026.00 = 3078.00 -> Quad 3, raw 6 -> 1017(.00)
sample = -32765, expected = -32765

10. test FAST_FORWARD functionality: double speed step
take one step in fast forward
Due to pipeline, hardware adds to the hidden advanced state: 2949.75
expected_addr = 2949.75 + 256.50 = 3206.25 -> Quad 4, raw 134 -> 889(.00)
sample = -32067, expected = -32067

11. test FAST_FORWARD functionality: continuous sine wave
take eight steps in fast forward
expected_addr = 3206.25 + 2052.00 = 5258.25 (wrap to 1162.25) -> Quad 2, raw 138 -> 885(.00)
sample = 32025, expected = 32025

No errors!
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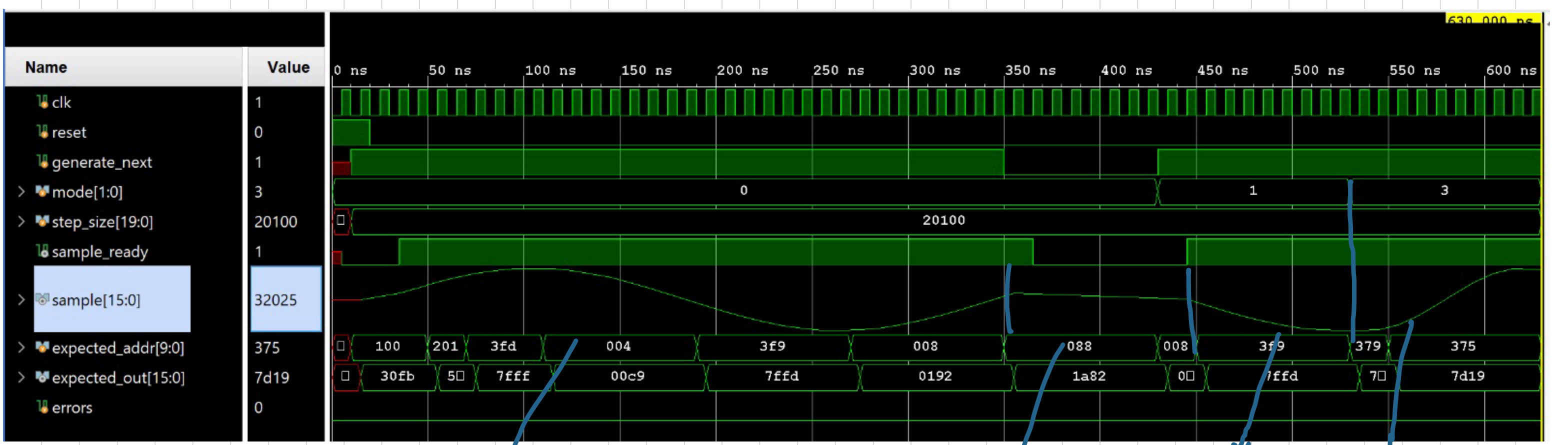


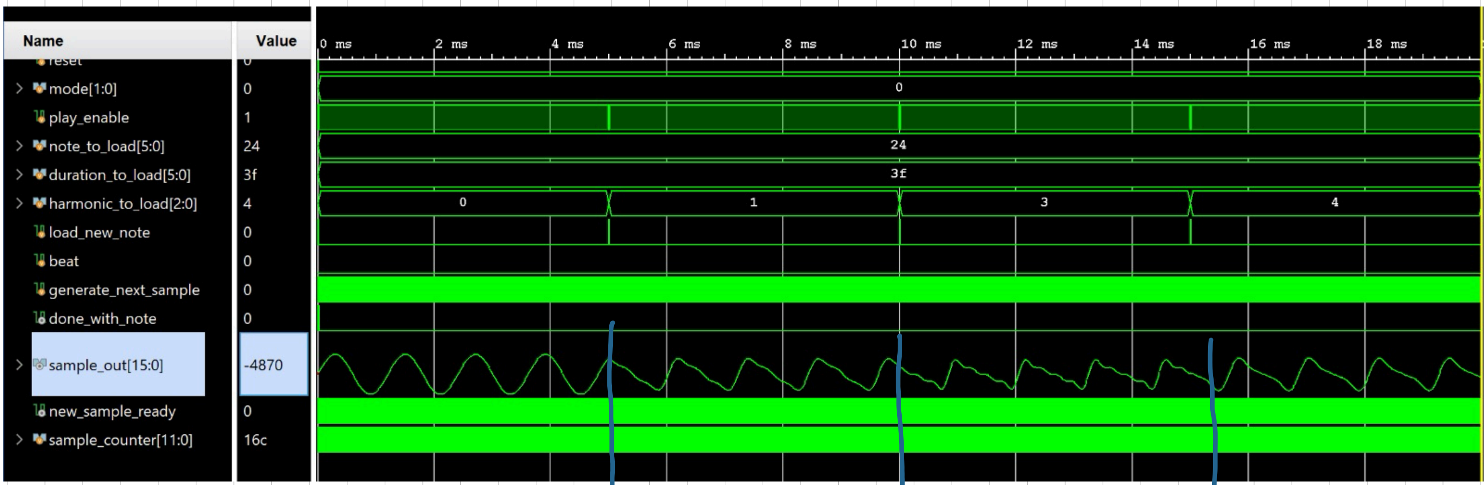
perfect forward sine wave.

paused

rewind at the same sample

fast forward at the same sample

harmonic_tb.v → plays different harmonics



harmonic 0 | 1 | 3 | 4